Determining Location and Movement Pattern Using Anonymized WiFi Access Point BSSID

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***Abstract*— Location tracking applications are usually based on exact GPS location information and non-anonymized WiFi Access Point (AP) information. It has been assumed that anonymizing WiFi AP information is an effective way to preserve location privacy of the connected users. In this work, our goal is to show that the privacy of connected users' movement pattern can still be compromised by a determined attacker even if the WiFi AP information is anonymized. We investigated the feasibility of tracking user's movement between locations from anonymized WiFi AP BSSIDs based on large Device Analyzer datasets. Our experiments show that the user's daily movement pattern can be identified with almost 83% accuracy.**

Keywords-location; movement; privacy; anonymized; WiFi

I. INTRODUCTION

Location tracking has become a common practice these days as location based services continue to gain popularity. A prominent example is the Google Maps service, which can track a user continuously while the user's device remains on and the tracking result can be easily viewed at [1]. Similarly, Facebook recently started a "Nearby Friends" feature [2] which tracks each user to find nearby friends. These are just two well-known instances of location tracking out of hundreds of other similar applications available in the Google Play, Apple App Store, and Windows Store. Most of these applications operate by collecting GPS coordinates directly from the smartphones. A few of them alternatively use WiFi AP information to locate a user when there is no GPS service available. They do it by correlating the WiFi AP information with the available GPS information and storing it into a database. However, all these applications need the actual values of GPS co-ordinates or WiFi AP BSSIDs to identify exact or approximate locations. Now, due to the concern for location privacy, users may prefer not to allow accessing GPS information but WiFi AP information might be readily accessible.

In this work, we investigated how much location and movement information can be extracted from nearby WiFi AP BSSIDs. We used a portion of the Device Analyzer [3] dataset of 100 billion records from 17,000 devices, where the dataset included periodic scanning and discovery information of nearby WiFi APs for each device. The scanning data included WiFi BSSIDs that are anonymized; that is, they are hashed to conceal actual values. The dataset did not include any GPS location information. Therefore, we had no way to determine the exact location of the user at any given time. However, the dataset included hashed GSM or CDMA cell location information, which gave us a way to verify the correctness of our findings using hashed WiFi AP information.

We developed an algorithm based on the WiFi scanning data, which is available every 5 minutes in most of the cases. Our algorithm can identify the visiting times and the duration of stays for places frequently visited by an individual. We have developed another algorithm that can do the same using GSM or CDMA cell location information. We have compared both results for 366 devices and found almost 83% matches on average. Moreover, we can deduce some important personal information for an individual from these patterns. For example, the places visited most frequently during night times can most likely be assumed as the home location of an individual. Similarly, the places visited most frequently during day time on weekdays can most likely be assumed as the work location.

The remainder of this paper is organized as follows. In Section II, we give an overview of related work. In Section III, we describe the dataset used in our work. In Section IV, we present our method and algorithm. In Section V, we discuss the results of our evaluation tests. Finally, we draw conclusions in Section VI.

II. RELATED WORK

WiFi positioning system [4] is not new in the industry. Google, Skyhook Wireless and others have WiFi SSID databases which can be used to find location of a given SSID. The difference between their work and ours is that they have actual GPS coordinates of the SSIDs, which they can use to geo-locate the user. Our work is based on hashed BSSIDs collected from Device Analyzer dataset where no information about the location is available. Few similar works exist in the academia utilizing actual location of the user. Montoliu et al. [5] proposed a client-server based system which records data about GSM, GPS, Bluetooth, WiFi, motion, as well as most of the smart phone applications. They used the data collected from these sensors to learn places of interest. Choujaa et al. [6] used cell tower IDs to record users' location, and used these locations to

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predict users' movement pattern. They showed how to select time points to reduce uncertainty on a mobile phone users' activity at a given time of the day and up to three weeks in the future. They also quantified how the user's activity at a certain time of the day can be determined from another user's activity at another time. Do et al. [7] studied the place labeling problem in a location privacy-sensitive setting when by design a physical geo-location is not available as a feature. They addressed the following questions using automatic location labeling method: How many places do people go to in everyday life? How often are these places visited? What are the effects of demographics and calendar in the dynamics of place visits? However, true location collected from GPS and WiFi were available to them. Zhang et al. [8] found that a large portion of visited locations is missing from check-ins, and most check-in events are either forged or superfluous events by comparing GPS traces against Foursquare check-ins. However, they also had access to the exact location from GPS as ground truth.

III. Description of Dataset

The dataset we use in this research is obtained from the Device Analyzer project [3] at University of Cambridge. Device Analyzer is an app developed to work with Android devices running Android 2.1 or higher versions, and is available for download by interested users. Currently, there are 21,432 active contributors to this project globally.

The complete Device Analyzer dataset contains 100 billion records collected from 17,000 devices during past few years. The data associated with each device is recorded for varying duration ranging from a few weeks to over a year. We selected a portion of this dataset which consists of 366 devices - all containing records collected for at least six months. Each record contains timing information and one specific key value pair. A detailed description of all the key value pairs can be found in [9]. We only used a few key value pairs related to WiFi and cell-location. For WiFi, we used "scancomplete" and "BSSID". "scancomplete" indicates that a WiFi scan is finished. The value of "scancomplete" contains the number of visible APs. The "BSSID" key under "scan" key has hashed value of the MAC addresses of different APs. For cell-location, we used "cid" in case of GSM signals and "basestationid" in case of CDMA signals. "cid" is the Cell ID for GSM signals, while "basestationid" is the base station ID for CDMA signals. All of these values in the dataset have been hashed to conceal identity. Beside these keys, there is only one other key that we needed occasionally – the "shutdown" key. The existence of this key indicates that the device is currently powered down.

IV. Method and Algrorithm

Our goal was to identify as much location and movement pattern information as possible from the WiFi scanning data. Therefore, first of all, we extracted a timeline of visited places for the entire period of time when the scanning data was collected. As mentioned before, we had WiFi scanning data with 5-minute intervals for most of the entire period of time with only occasional gaps in time. Hence, we developed the timeline with 10-minute intervals starting from the very first occurrence of scanning data to the very last occurrence of the same. Since the exact location information is not available due to the anonymization of BSSID, we used 3-digit numerical IDs to distinguish between places. Thus, "001" should be the first place we identify. Places are identified by creating unique sets of WiFi AP BSSIDs. That means, if two scanned set of AP BSSIDs have at least one match between them, then they will be merged to form a larger set of APs and will be labeled with a unique place ID.

We develop this list of places with corresponding unique set of AP BSSIDs as we parse the dataset for a single device. Each time we encounter a "scancomplete" record, we start saving the set of AP BSSIDs. Once we have the whole set for a single "scancomplete" instance, we check with all the previous sets to find a match, i.e. a previous set of APs with at least one overlapping BSSID. If we can find one such instance, the new set is merged to the old set. Otherwise, we create a new set of AP BSSIDs and label it with the next place ID, which is incremented every time by 1. In this way, we construct the whole timeline with corresponding places. The algorithm to construct the timeline is shown in Figure 1.

Now, from the timeline, we can construct a daily timeline or "dayline". A "dayline" is where we have the list of most frequently visited places for every 10 minute interval of a single day. From dayline, we can infer the place ID that is most likely to be the user's home location. The approach is to identify the most frequently visited place from 12 AM to 8 PM after adjusting for the time zone. Similarly, by only considering the weekdays while constructing the dayline, we can infer the place ID that is most likely to be user's work location by identifying the most frequently visited place from 9 AM to 5 PM after adjusting for the time zone. Here, it should be mentioned that due to the anonymized nature of the Device Analyzer dataset we were unable to actually verify these claims. However, we found a way to verify the correctness of the dayline pattern by constructing another dayline using cell-location information.

The cell-location information is not as regular as the WiFi scanning information. The intervals between two consecutive cell ID records vary greatly. Therefore, we used the change in hashed cell ID values to distinguish in subsequent records to indicate the change in location. In a similar fashion, we labeled the places with 3-digit unique incrementing numbers. That means, as long as the records indicate the existence of the same hashed cell ID value, we assume that the user is staying in the same place and not moving. The only exception is when we encounter a "shutdown" record. In this case, there will be a time gap, since the device was turned off and we have no way of knowing where it was during that period of time. Once the cell ID value changes to a new one, we record it as a movement to a new place. In this way, the number of total places a user visits should be equal to the number of unique cell ID values encountered by the device during the whole time period. We construct the timeline and dayline exactly the same way with 10-minute intervals.

```
Algorithm: ExtractWifiLocation
Input: wifi scanning data for a device
Output: a list of (time,place) tuples

previoustime = NULL
scanning = FALSE
wifiplaces = dict()

for each <time, event, bssid> in data
     if event is scancomplete then
          if scanning is TRUE then
               currentplace = NULL
               if wifipalces is EMPTY then
                    insert newplace into wifiplaces
                    currentplace = newplace
               else
                    for each place in wifiplaces
                         match = calculatematch(newplace.aplist, place.aplist)
                         if match > 0 then
                              merge(place.aplist, newplace.aplist)
                              currentplace = place
                              break
                         end if
                    end for
                    if no match then
                         insert newplace into wifiplaces
                         currentplace = newplace
                    end if
               end if
               print(previoustime,currentplace)
          create newplace
          previoustime = time
          scanning = TRUE
     else if previoustime == time
          if event is scan then
               append bssid to newplace.aplist
          end if
     end if
end for
```

Figure 1. Algorithm to construct the WiFi timeline.

V. RESULT AND ANALYSIS

We constructed the dayline using both WiFi AP and cell-location scanning data for 366 devices. Then, we constructed a 0-1 movement pattern for both of the methods, where a "0" indicates no change of place and a "1" indicates movement. Then, we compared the patterns generated by both of the methods for all 366 devices. The results indicated almost 83% match on average. This means, even though we cannot identify the exact locations, we can construct a daily movement pattern for an individual with considerable accuracy. Figure 2(a) shows the most frequently visited places as percentages of the whole day constructed from a WiFi dayline. Figure 2(b) shows the percentages of frequently visited places during night time for the same dayline as in Figure 2(a). Figure 2(c) shows a similar WiFi dayline chart as Figure 2(a) for the same device, but only considering weekdays. Figure 2(d) shows the percentages of frequently visited places during daytime for the dayline of Figure 2(c).

From the charts above, it is clearly visible which place IDs are most frequently visited on a certain time period. In case of Figure 2(b), we can say that Place-299 is the most likely candidate for home location. Similarly, in case of Figure 2(d), Place-295 can be considered as the most likely candidate for work location.

In addition, we performed experiments to verify our claim based on ground-truth information. We used one of our android smartphones to simultaneously record data using the Device Analyzer app and Moves [10] app. Moves is an app which automatically records the smartphone owner's walking, cycling, and running activities by collecting actual GPS location information. We collected the movement and location information using both of the apps for one month. Then, we ran the same algorithm on the Device Analyzer data and compared it to Moves data. We found 87% match between the daylines constructed using both approach. Fig. 3 shows a comparison between the most visited places recorded by both systems. Fig. 3(a) and 3(b) shows the results obtained from Device Analyzer data, while Fig. 3(c) and 3(d) shows results obtained from Moves data. For the Moves data, we replaced the actual GPS information with "Home", "Workplace", etc. Fig. 3(a) and 3(c) shows the most visited places during night time. From these two charts, we can clearly correlate "Place-034" with "Home". Again, Fig. 3(b) and 3(d) shows the most visited places during daytime considering only weekdays. From these two charts, we can clearly correlate between "Place-001" and "Workplace". This means, even if we cannot identify the actual location of an individual from the anonymized data, we can still correctly identify historical timeline of a person staying at home or at workplace.

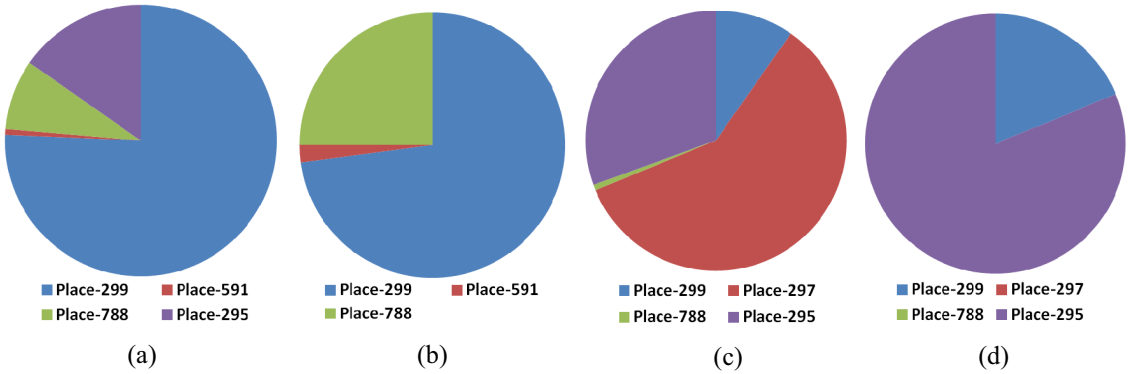

Figure 2. Distribution of visiting frequencies for places in different settings.

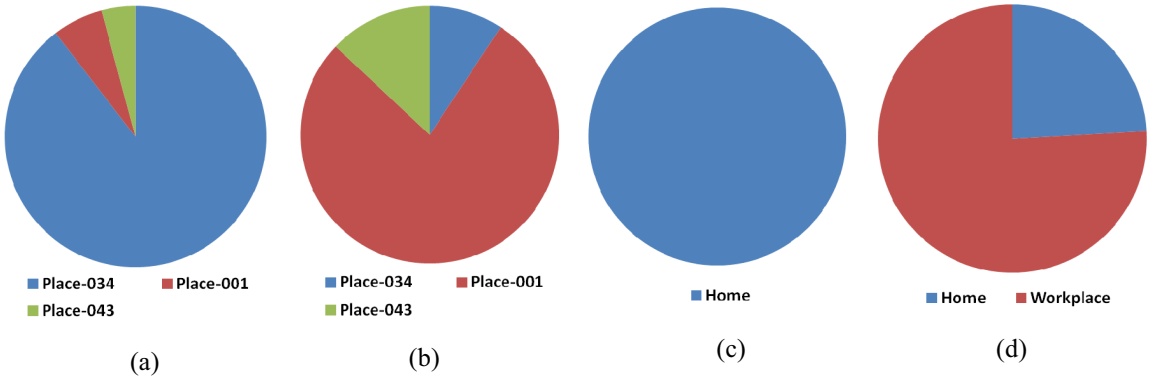

Figure 3. Results of the ground-truth experiment.

VI. Conclusion

In this work, we explored the feasibility of using WiFi AP BSSIDs to identify location and movement patterns for an individual smartphone user. Our experiments show that we can extract useful location and movement information to some extent even if the BSSID data values are anonymized. In addition, we presented an alternate approach to verify the accuracy of this method using smartphone cell-location information. Our results indicate that hash-based anonymization of WiFi or other location-variant data is not sufficient to completely prevent location or movement tracking.

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